

Title: Drones & UAV Technology

Project Name: SkyGuard

Team Name: Chaotic Bots

Members Names:

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** The following report is for the project “SkyGaurd” that is made by the team Chaotic Bots consisting of Akshay Ajit Bhawar , Revanth Sriraj , Joshua John & Srinidhi M Students of 2th year 4th semester. The report looks over the technical and non technical aspects of the Project with applications, limitations and over coming limitations. **

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Project Summery

SkyGuard, a brainchild of team Chotic Bots, emerges as a pioneering solution positioned to address the multifaceted challenges encountered by farmers, industrial workers, first responders, and security agencies alike. This cutting-edge attachment offers unparalleled adaptability, seamlessly integrating with off-the-shelf drones or vehicles, thereby enhancing their capabilities for diverse applications.

At its core, SkyGuard features a single-board computer responsible for onboard processing and data collection. This computing powerhouse enables real-time analysis of sensor data and facilitates swift decision-making. Coupled with an onboard GPS module, SkyGuard effortlessly acquires precise coordinates, crucial for navigation and location-based tasks.

However, what sets SkyGuard apart is its resilience in adverse conditions. Equipped with transceiver modules, it ensures seamless communication even in remote areas with no cellular or internet connectivity. Leveraging radio communication with extended range capabilities, SkyGuard remains steadfast in the face of challenging environments, offering uninterrupted surveillance and data transmission.

Moreover, SkyGuard is designed with user customization in mind. Utilizing advanced machine learning techniques, users can tailor the device to their specific requirements and operational settings. Whether deployed in military border patrols, search and rescue missions, or industrial inspections, SkyGuard empowers users to adapt its functionalities to suit their unique needs.

In military scenarios, SkyGuard proves invaluable for border surveillance, offering continuous monitoring and detection of illegal crossings with unprecedented efficiency. Similarly, in search and rescue operations, its rapid deployment and versatile capabilities enable swift response and coordination, potentially saving lives in critical situations.

In essence, SkyGuard epitomizes innovation and versatility, heralding a new era of aerial surveillance and exploration. With its ability to transcend limitations and adapt to diverse environments and applications, SkyGuard emerges as a game-changer, empowering users across various sectors to enhance safety, security, and efficiency in their operations.

Explanation using application

Military Border Patrol: SkyGuard enables continuous surveillance along sensitive borders, such as the India-Bangladesh border, detecting illegal crossings and smuggling activities with precision and efficiency. Its adaptability and extended operational capacity offer a significant advantage over traditional patrols, enhancing border security and minimizing illicit activities.

Search and Rescue Operations: In search and rescue missions, SkyGuard's rapid deployment and advanced capabilities facilitate swift response and coordination. By providing real-time aerial surveillance and precise location tracking, it enhances the effectiveness of rescue efforts, potentially saving lives in critical situations.

Abstract

SkyGuard brings a new era in aerial surveillance and exploration, blending advanced technologies to meet the dynamic demands of industrial and First Responders domains. Anchored by a sophisticated array of hardware, including the Raspberry Pi 4 with Camera Module, ESP32 Microcontroller with GPS Sensor, and LoRa Transceiver Modules with the Ardupilot Flight Controller and RC Controller, SkyGuard revolutionizes real-time person detection, precise location tracking, and long-range wireless communication.

At its core, the Raspberry Pi 4 harnesses computational power and versatility to capture and process high-resolution imagery, fueling SkyGuard's unparalleled person detection capabilities. Enhanced by TensorFlow lite machine learning algorithms, SkyGuard achieves exceptional accuracy in identifying and tracking individuals, elevating surveillance and security paradigms to new heights.

Using the Raspberry Pi 4, the ESP32 Microcontroller, coupled with GPS precision, facilitates seamless transmission of target coordinates to remote monitoring stations across different location in lora rangewith 433mhz. Its low power consumption and wireless agility ensure uninterrupted surveillance and response coordination across diverse environments, reinforcing SkyGuard's adaptability and reliability.

Moreover, LoRa Transceiver Modules extend SkyGuard's reach beyond conventional communication bounds, enabling resilient long-range data transmission. This transformative capability unlocks opportunities in border surveillance, disaster management, and industrial inspections, positioning SkyGuard as a cornerstone of innovation in aerial surveillance and exploration.

In essence, SkyGuard embodies the convergence of cutting-edge technology and motivation to do good, poised to redefine industry standards and First responders horizons. As industries embrace the future of drone systems, SkyGuard stands as a beacon of excellence, ready to propel surveillance and exploration endeavors to unprecedented levels of efficiency and efficacy.

Introduction

In an era of rapid technological advancement, SkyGuard emerges as a beacon of innovation, combining autonomous drone capabilities with versatile attachments to redefine aerial exploration and surveillance. This project embodies the fusion of cutting-edge hardware and software, promising transformative solutions for diverse industries and applications. the two main components of the drone is stated below

Autonomous Drone:

SkyGuard's autonomous drone system represents a pinnacle of technological innovation, with its core foundation resting upon the Raspberry Pi 4 and the Ardupilot Flight Controller. The Raspberry Pi 4, renowned for its versatility and computational power, serves as the brain of the drone, orchestrating complex flight maneuvers and processing sensor data in real-time. Working in tandem with the Ardupilot Flight Controller, SkyGuard achieves precise navigation and flight control, essential for executing predefined flight paths and responding to dynamic environmental conditions.

Central to SkyGuard's autonomous capabilities are its advanced sensors and machine learning algorithms. Equipped with state-of-the-art sensors such as gyroscopes, accelerometers, and GPS modules, the drone gathers crucial data about its surroundings, including altitude, orientation, and geographic coordinates. This sensor data is then processed through sophisticated machine learning algorithms, enabling SkyGuard to autonomously identify and track individuals in real-time with unparalleled accuracy and efficiency. Such capabilities revolutionize traditional surveillance and security operations, offering enhanced situational awareness and response capabilities across various industries.

Attachments:

SkyGuard's versatility is further augmented by its modular attachment system, enabling seamless integration of specialized tools and sensors tailored to specific applications. Equipped with single-board computers and LoRa transceiver modules, SkyGuard's attachments enable real-time data collection and transmission, even in remote or challenging environments where traditional communication methods may falter. The single-board computers provide computational power and data processing capabilities, allowing for onboard analysis and decision-making. Meanwhile, the LoRa transceiver modules facilitate long-range wireless communication, ensuring robust connectivity between SkyGuard and ground control stations or remote monitoring centers.

Whether deployed for border patrol, search and rescue missions, or environmental monitoring, SkyGuard's attachments offer limitless possibilities for innovation and problem-solving. From thermal imaging cameras for nocturnal surveillance to environmental sensors for pollution monitoring, these attachments empower users to tailor SkyGuard to meet the specific needs of their applications, making it an indispensable asset across industries.

Motivation

While watching the "Deadliest Catch" TV show, I couldn't help but feel deeply moved by the challenges faced by coast guards and first responders. The sheer difficulty of locating individuals lost at sea amidst harsh conditions struck a chord within me. Waiting for rescue, often enduring agonizingly long periods, seemed incredibly tough. It dawned on me just how daunting search and rescue missions can be, especially in vast and unforgiving environments like oceans, forests, and remote areas.

Similarly, documentaries depicting soldiers patrolling borders left a lasting impression. Their relentless efforts in braving treacherous terrains and harsh climates showcased the immense sacrifices made to protect borders. Witnessing their arduous journeys filled me with empathy and a strong desire to make a difference.

These experiences sparked the inception of SkyGuard—a project driven by a sincere desire to alleviate the hardships faced by those on the frontlines of rescue and border protection. SkyGuard isn't just another project; it's a beacon of hope and innovation. By harnessing the power of computer vision technology, SkyGuard aims to revolutionize search and rescue operations, making it easier and quicker to locate individuals in distress.

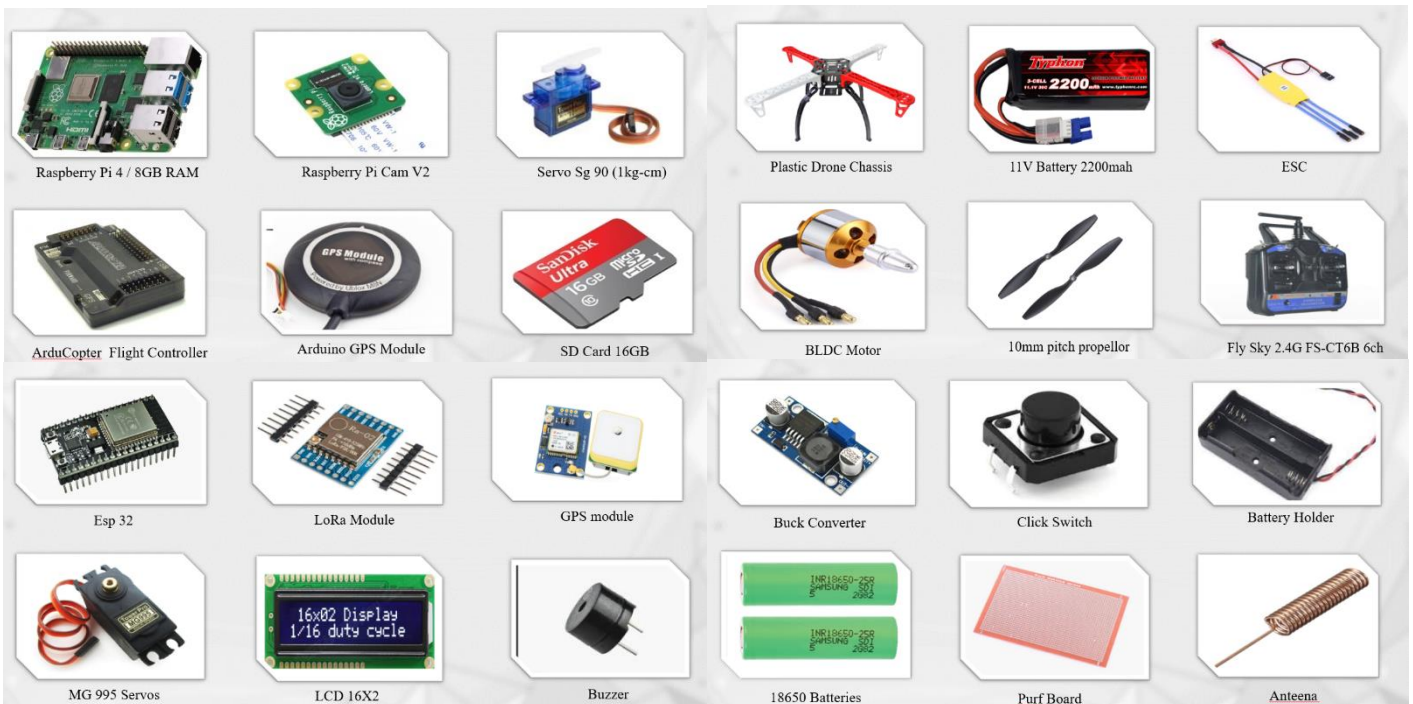
Moreover, SkyGuard's surveillance capabilities seek to enhance border security, ensuring the safety of all involved, from coast guards to border patrol agents. The project embodies a commitment to offering a lifeline of support to those who selflessly dedicate themselves to safeguarding others.

With its user-friendly interface and advanced technology, SkyGuard promises to streamline rescue efforts and reduce critical response times. By providing real-time data and precise location tracking, it empowers rescue teams to navigate challenging environments more effectively.

In essence, SkyGuard is more than just a project—it's a testament to human ingenuity and compassion. With each innovation, it strives to make the world a safer place for everyone, one rescue mission at a time.

Components Used

1. Raspberry Pi 4 / 8GB RAM
2. Raspberry Pi Cam V2
3. Servo Sg 90 (1kg-cm)
4. ArduCopter Flight Controller
5. Arduino GPS Module
6. SD Card 16GB
7. Plastic Drone Chassis
8. 11V Battery 2200mah
9. ESC
10. BLDC Motor
11. 10mm pitch propeller
12. Fly Sky 2.4G FS-CT6B 6ch
13. Esp 32
14. LoRa Module
15. GPS module
16. MG 995 Servos
17. LCD 16X2
18. Buzzer
19. Buck Converter
20. Click Switch
21. Battery Holder
22. 18650 Batteries
23. Purf Board



Design Construction

The journey of designing SkyGuard commenced with a thorough analysis of the challenges faced in search and rescue missions and border security. Recognizing the critical need for a solution, our team embarked on a brainstorming session to craft the design of SkyGuard.

During this phase, we explored numerous ideas to address the complexities of the problem statements, aiming for effectiveness and simplicity. Valuable guidance from retired ISRO expert, "Anil Sir," with advise “the best design is the one which solves complex issues in the simplest way” this underscored the importance of simplicity in design.

Guided by Anil Sir's wisdom, we honed our focus on refining SkyGuard's design to ensure it was both robust and user-friendly. Our objective was to create a product that could be easily produced and utilized without compromising strength.

Through 15 iterative cycles, we diligently incorporated feedback and made improvements, drawing closer to our goal with each iteration. Continuous refinement led us to the final design of SkyGuard.

In the final stage, we utilized Fusion 360 to create a software model, enabling us to assess its practicality and functionality through analysis. This digital approach allowed us to simulate real-world conditions, ensuring that SkyGuard met the stringent requirements of its intended applications.

The following pages will show the Contant:

Fig 1.1 : Rendered Image of the SkyGuard Drone

Fig 1.2 : Rendered Image of the Attachment

Fig 1.3 : Rendered Image of the Receiver

Fig 2.1 : Sketch Image of the SkyGuard Drone

Fig 2.2 : Sketch Image of the Attachment

Fig 2.3 : Sketch Image of the Receiver

Fig 3.1 : Dimensional Rendered Image of the SkyGuard Drone

Design Construction

****After 15 iterations of our project we came to this design which solves all our requirements and outperforms other designs****

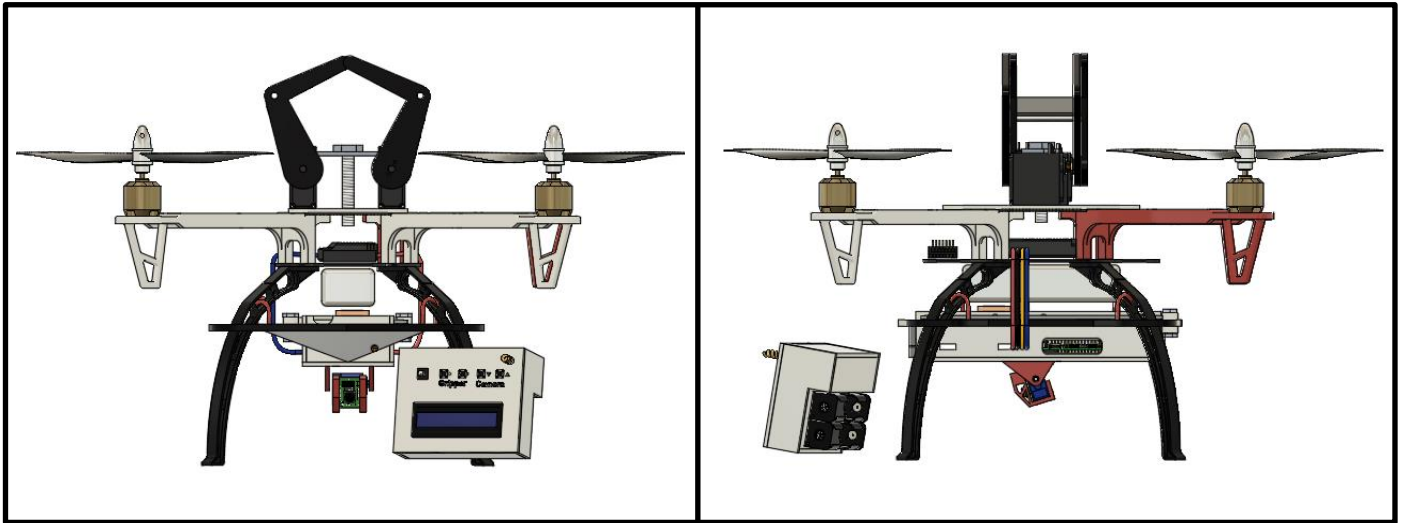


Fig 1.1 SkyGuard Design

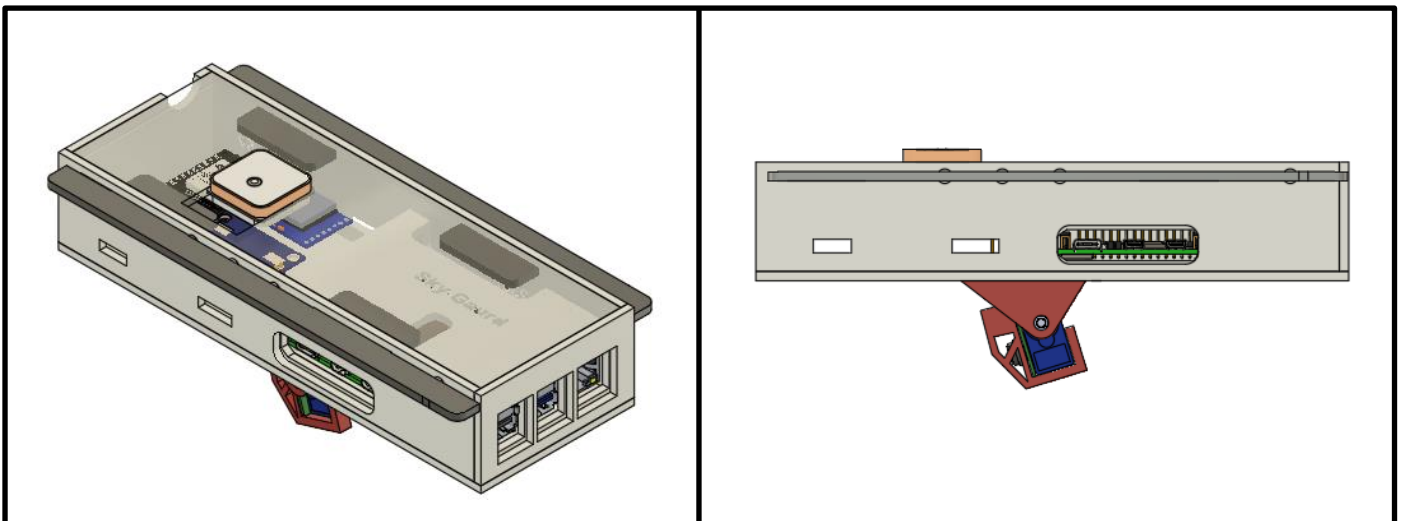


Fig 1.2 Attachment Design

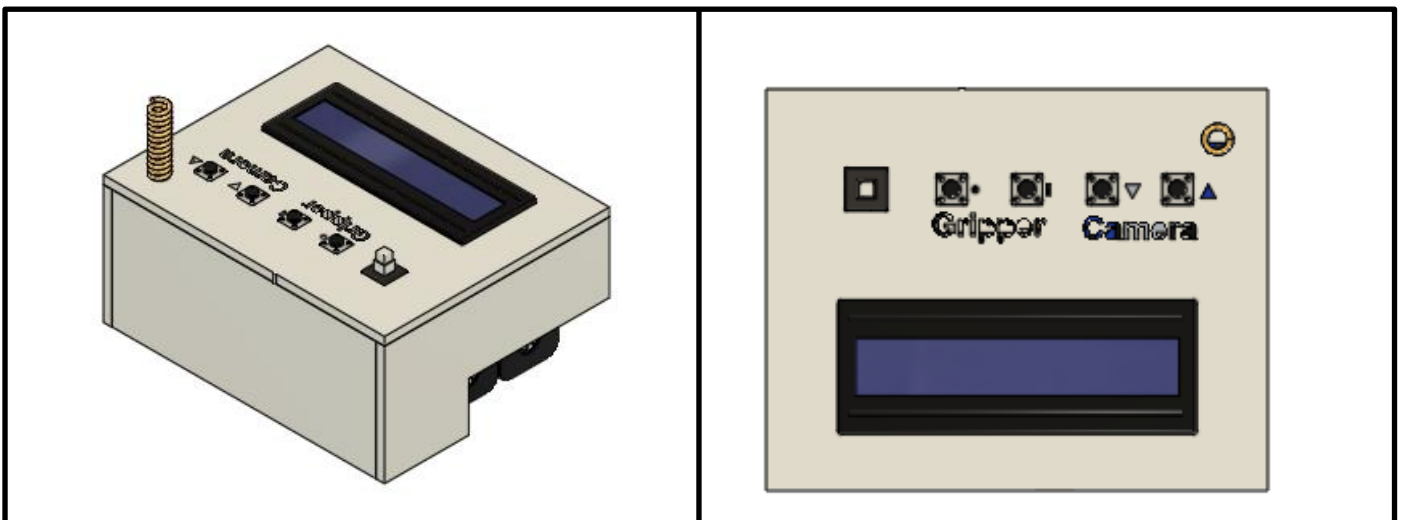


Fig 1.3 Receiver Design

Design Construction

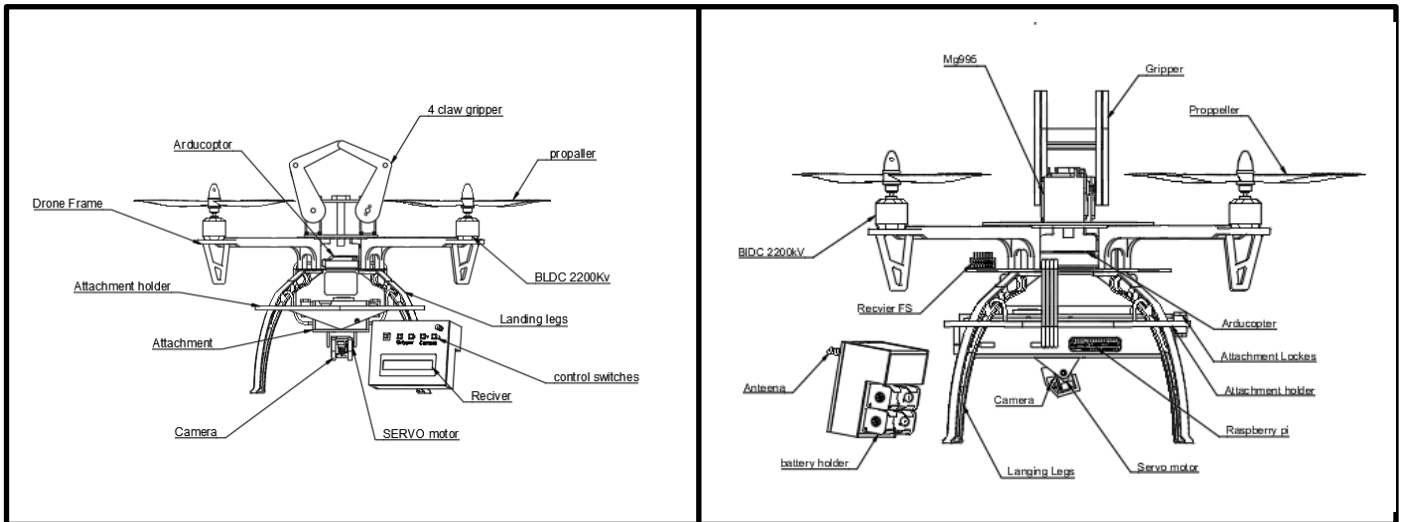


Fig 2.1 SkyGuard Design

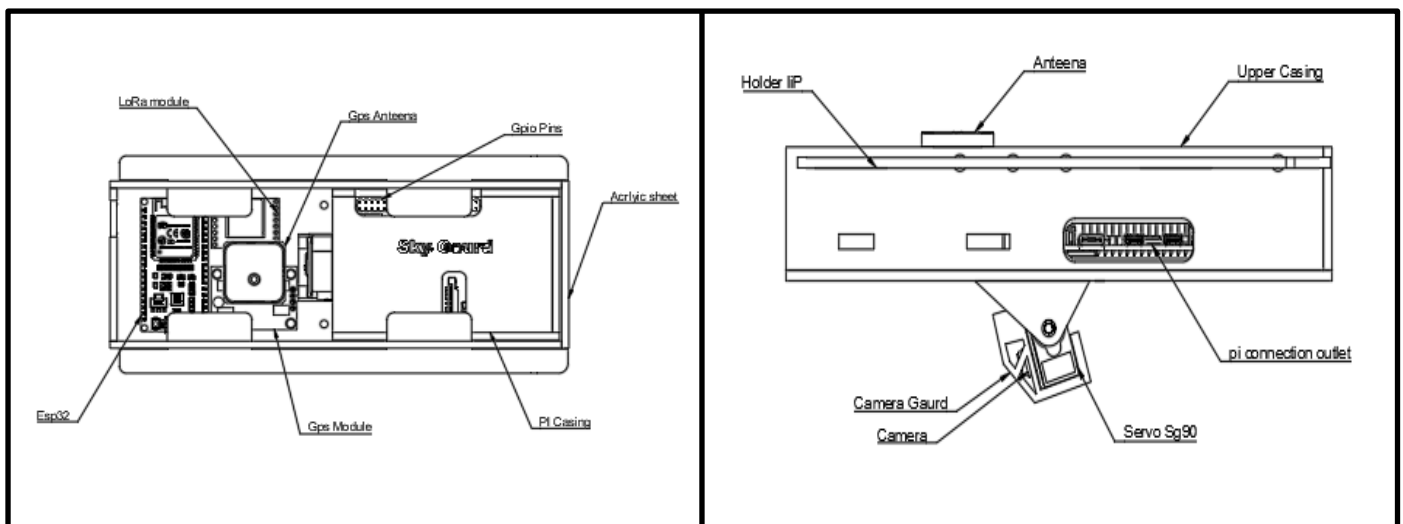


Fig 2.2 Attachment Design

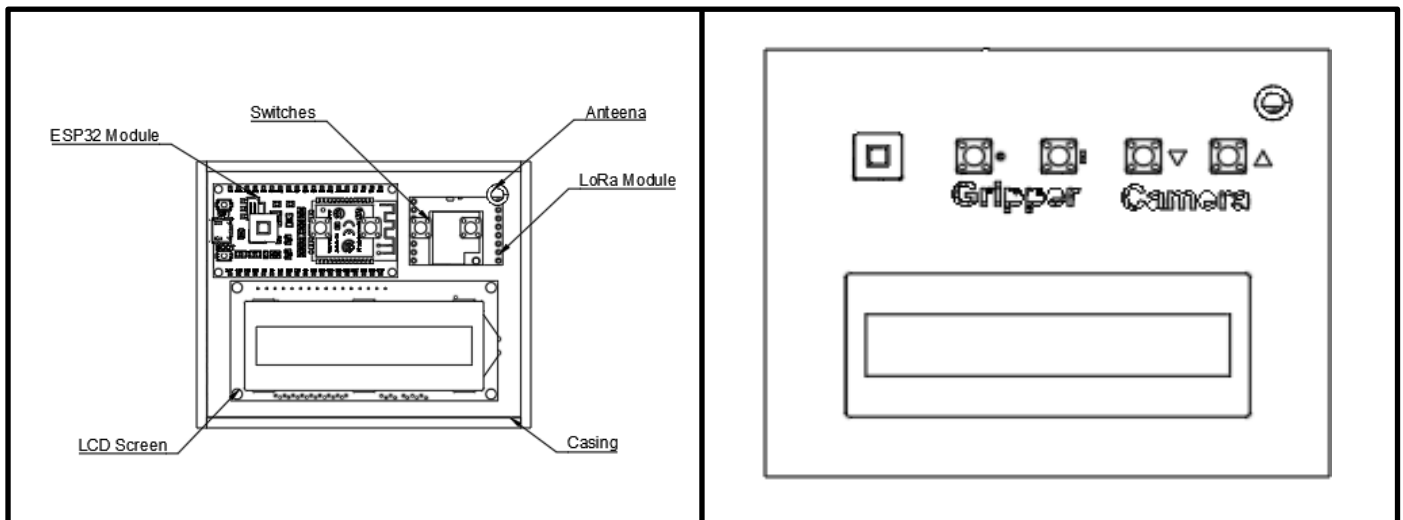


Fig 2.3 Receiver Design

Design Construction

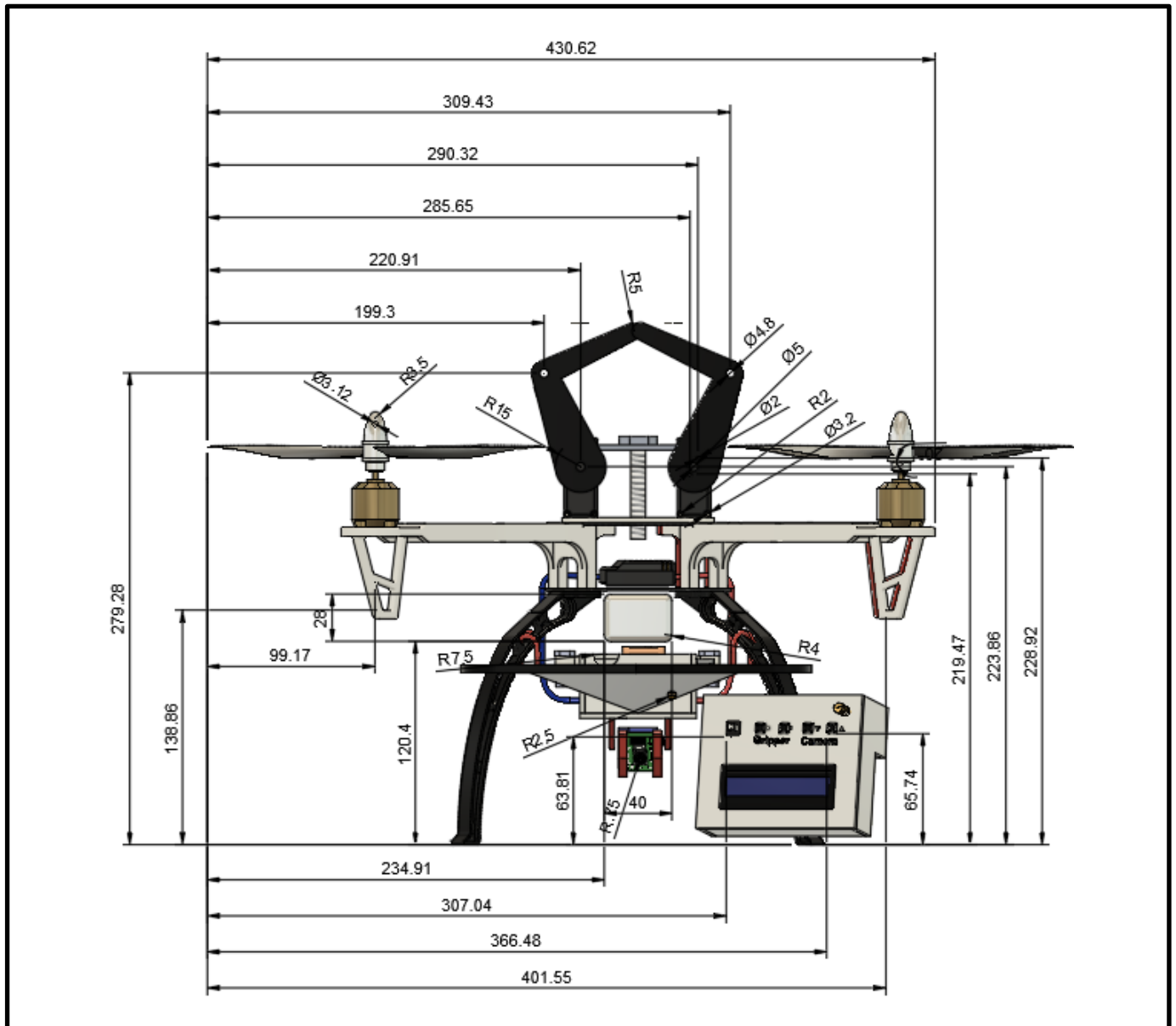


Fig 3.1: Dimensional Render

Specification of construction:

Made up of acrylic sheets :

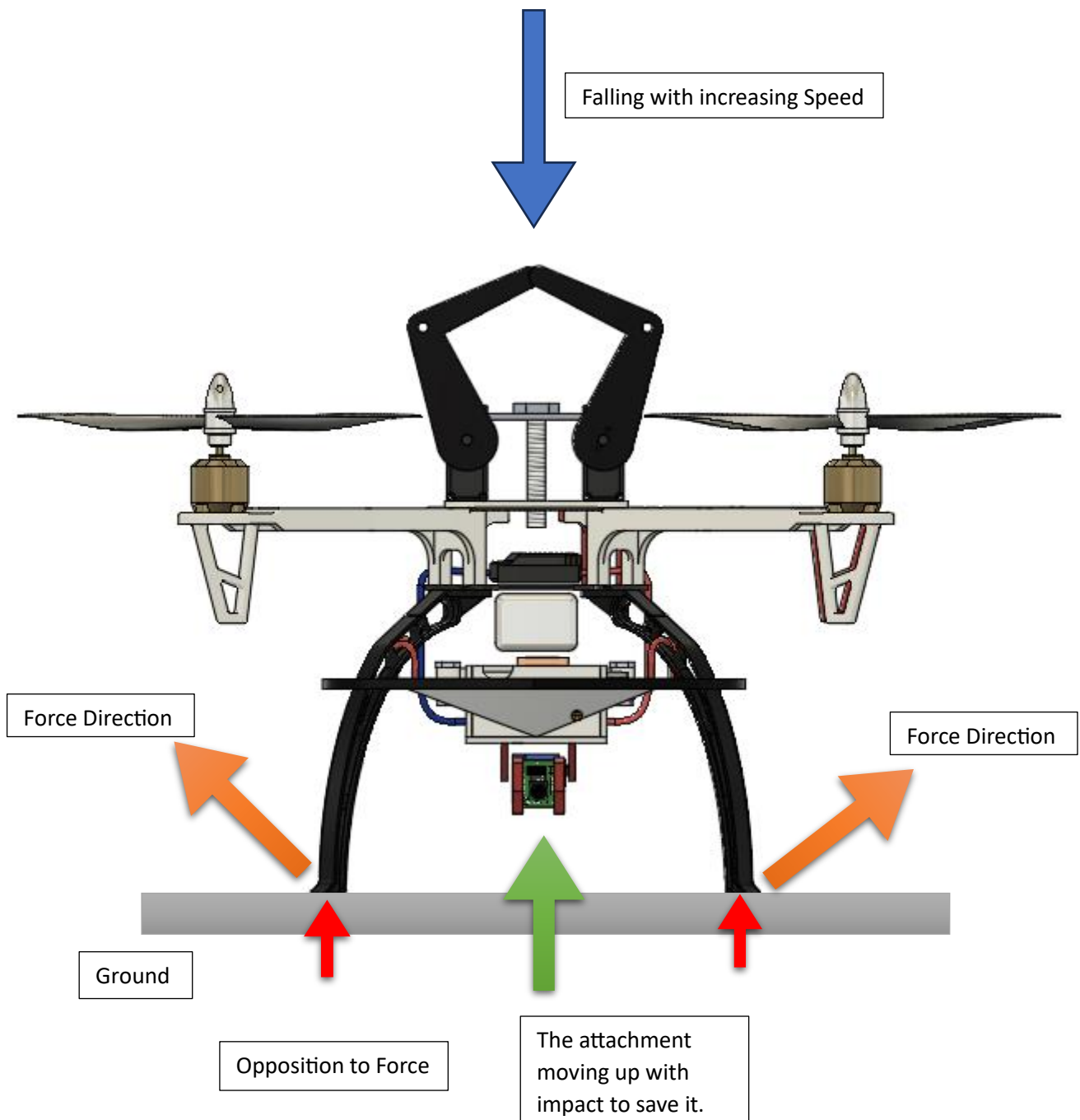
5mm

white, 2mm transparent , 5mm black

Design Analysis

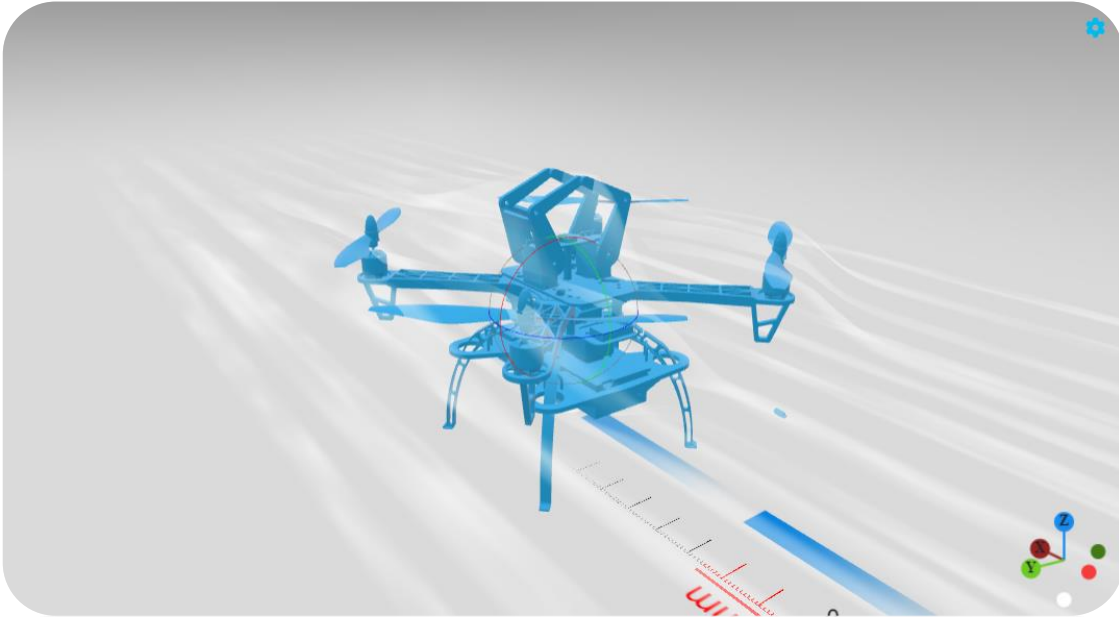
The advantage of the design during impact or free fall toward the ground

our project mentor Anil Kumar sir who is Retired ISRO employ gave us an advise that stayed with use as the working principal “The best design is the one, which can solve complex problem in simple ways”. Keeping this in mind we made the holder design and attachment design. We where not able to analyse it in a software because of low computational power of our systems

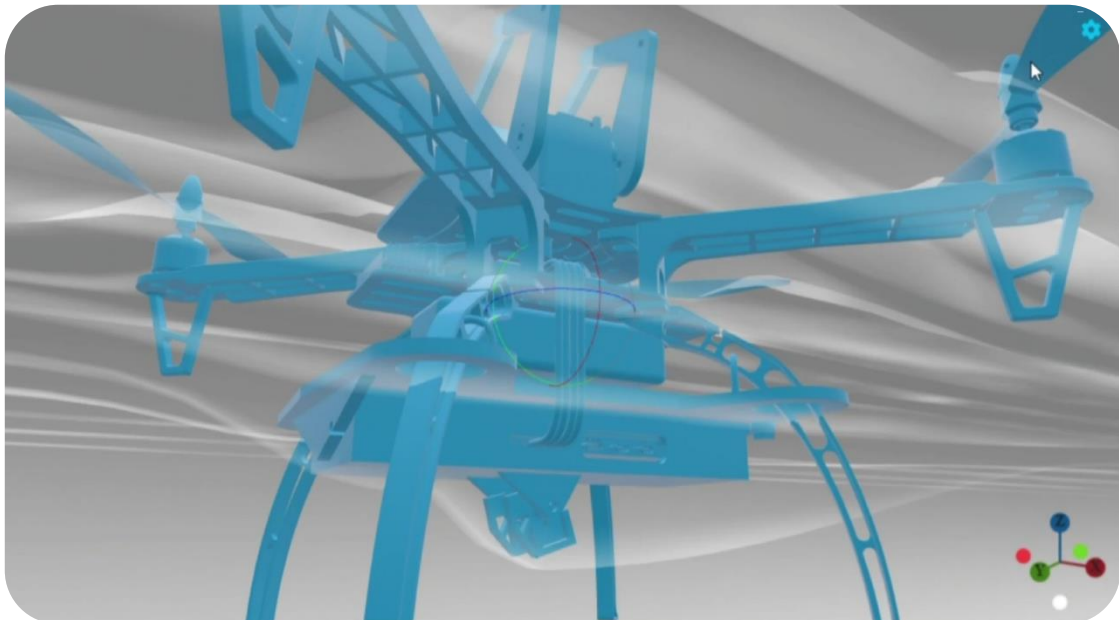


We have done impact testing and the drone legs and it performs well under the conditions. During the legs spread in opposite ways this cause the attachment to be pushed up indirectly saving it from ground impact and also saves the camera from any damage. This reduces the forces that will be experienced by the attachment & lipo battery which is important for safety

Aerodynamic Analysis



After designing the analysis of Stress and the Aerodynamics was carried out. The picture here shows the Aerodynamics of the drone. as we can see the gripper has produced minute drag due to its open design also the highest drag that the drone is experiencing is at the bottom near the attachment part and near the camera. As seen in the wave pattern. Because of a block based design in future we will try to make aerodynamic to reduce drag



Calculations were carried out by the software "SimScale" from which we get to know that the Drag in Newtons it was calculated to be approx. 2.2N if moving toward front at the angle of 90 degree (easy to compute). At the speed of 4m/s and battery with 3300mah and 12v. with all the accessories on board like attachment and 3d print pipes practically the drone moving forward will be at the tilt of 60 degree and therefore there will be increase drag

Methodology

SkyGuard operates as an integrated system designed to enhance the efficiency and effectiveness of surveillance and search and rescue missions, catering to the needs of first responders and military personnel, even in remote locations. The methodology involves a series of steps to deploy, operate, and utilize SkyGuard effectively:

1. Deployment Preparation: The user, whether a first responder or military personnel, prepares SkyGuard for deployment by setting waypoints for the drone's flight path. These waypoints can be predefined locations or manually adjusted as needed to cover specific areas of interest.

2. Autonomous Flight: Once deployed, SkyGuard autonomously follows the programmed flight path at the correct speed and altitude, ensuring comprehensive coverage of the designated area. The drone's flight controller maintains stability and precision throughout the operation.

3. Attachment Activation: While the drone is in flight, the user activates the attached sensors and cameras to scan the area below. These attachments are seamlessly integrated into SkyGuard's structure, enabling real-time data collection and transmission.

4. Person Detection: Using computer vision technology, SkyGuard scans the area for individuals in distress. By analyzing upper body, face, lower body, or arm features, the system detects the presence of a person within its surveillance range.

5. Location Acquisition: Upon successful detection, the attachment's Raspberry Pi 4 sends a signal to the ESP32, triggering the GPS Neo 6 module to acquire the longitude and latitude coordinates of the detected person's location.

6. Data Transmission: The acquired coordinates are transmitted via LoRa communication to a secondary LoRa module, which acts as a receiver. LoRa technology ensures reliable long-range transmission, even in remote or obstructed environments.

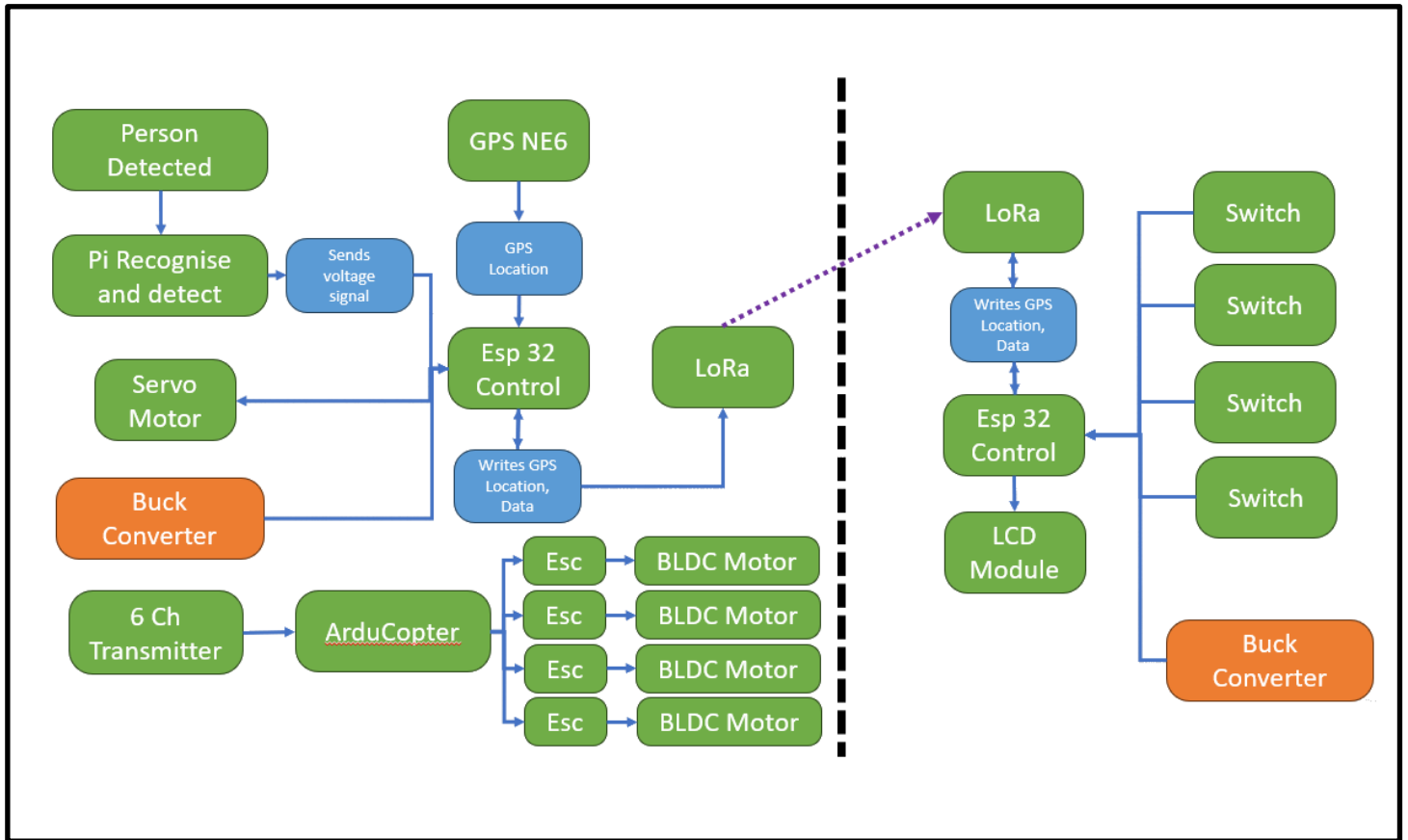
7. Alert and Display: The receiver module receives the transmitted coordinates and displays them on an LCD screen and on the app made for the user, providing real-time information to the user. Additionally, a buzzer sounds to alert the user of the detected person's location.

8. Attachment Control: The receiver module is equipped with switches to control attachment functionalities, such as adjusting camera angles or activating the gripper. These controls are also facilitated through the LoRa communication protocol, allowing for seamless operation.

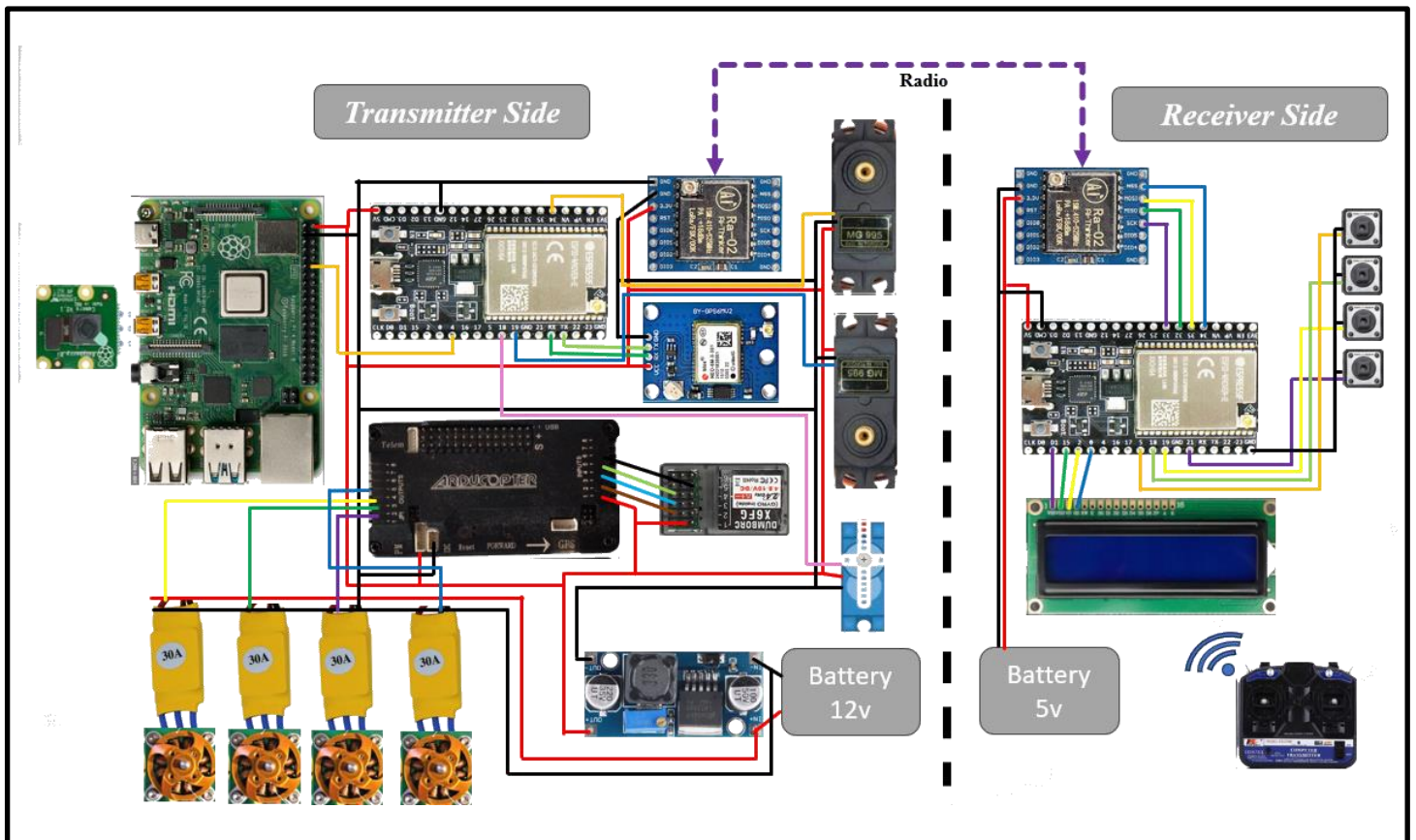
Summery SkyGuard Working:

SkyGuard operates by combining advanced technology, including computer vision, GPS, and LoRa communication, to autonomously scan and detect individuals in distress during surveillance or search and rescue missions. It begins with the user deploying the drone and setting waypoints for its flight path. As the drone flies, the attached sensors and cameras scan the area below, utilizing computer vision algorithms to identify individuals. Upon detection, the system triggers the acquisition of location coordinates, which are transmitted via LoRa communication to a receiver module. The receiver module displays the coordinates on an LCD screen and sounds an alert, while also providing control functionalities for attached devices. This seamless integration of hardware and software allows SkyGuard to effectively locate individuals in need, even in remote or challenging environments, facilitating prompt response and assistance from first responders or military personnel.

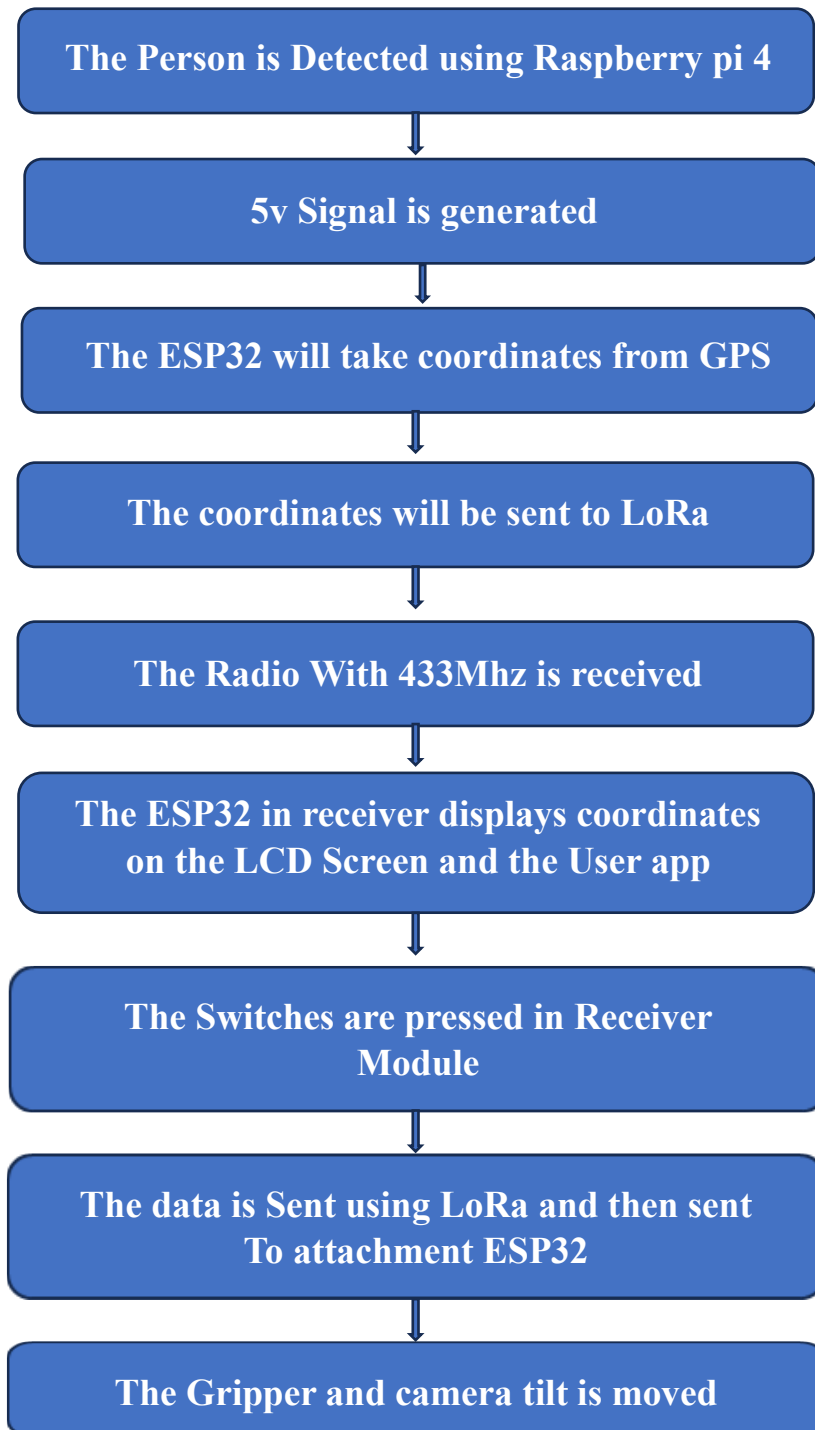
Block Diagram



Circuit Diagram

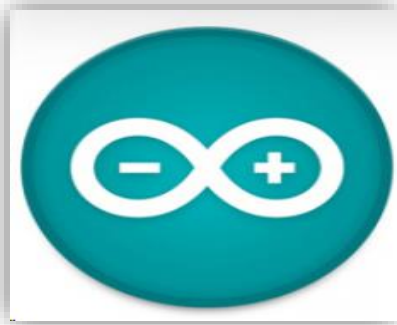


Flowchart

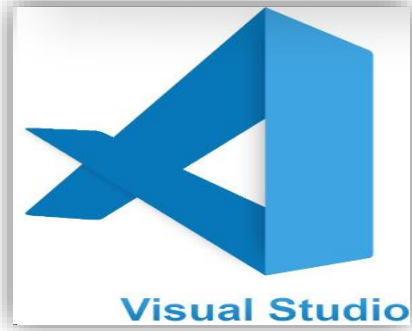


For other applications flow remains the same but the detection criteria changes with minute changes within the flow of the working. The model can be changed by the user and in future they can switch between trained models using a user app that will make the system modular and more user friendly

Software Dependencies



Arduino Ide



Visual Studio



Thunder Lazer



TensorFlow



Mission Planer



Fusion 360



Kodular

Coding

Arduino coding Attachment:

```
#include <SPI.h>
#include <LoRa.h>
#include <Wire.h>
#include <ESP32Servo.h>
Servo servo;
Servo servor;
Servo servol;

#define ss 5
#define rst 14
#define dio0 2

int pos = 45;
void setup()
{
  Serial.begin(9600);
  Serial.println("Attachment");
  pinMode(25, INPUT);
  pinMode(26, INPUT);
  LoRa.setPins(ss, rst, dio0);
  LoRa.begin(433E6);

  servo.attach(33);
  servol.attach(32);
  servor.attach(13);

  servo.write(pos);
  servor.write(180);
  servol.write(0);
}
```

```
void loop()
{
  int value = digitalRead(25);
  int value1 = digitalRead(26);
  int packetSize = LoRa.parsePacket();

  if (packetSize)
  {
    String received = LoRa.readString();
    Serial.println(received);

    if (received == "up")
    {
      Serial.println("Servo Up");
      pos = pos + 5;
      servo.write(pos);
      if(pos >= 90)
      {
        pos = 90;
      }
    }
    else if (received == "down")
    {
      Serial.println("Servo Down");
      pos = pos - 5;
      servo.write(pos);

      if(pos <= 0)
      {
        pos = 0;
      }
    }
    else if (received == "hold")
    {
      servor.write(70);
      servol.write(110);
    }

    else if (received == "leave")
    {
      servor.write(180);
      servol.write(0);
    }
  }
}
```

Coding

Arduino coding Receiver:

```
#include <SPI.h>
#include <LoRa.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define ss 5
#define rst 14
#define dio 2
String Storage;
String received;
String data1;
String data2;

void setup() {
  Serial.begin(9600);
  Serial.println("LoRa Receiver");
  lcd.begin();
  lcd.backlight();
  pinMode(12, OUTPUT);

  LoRa.setPins(ss, rst, dio);
  LoRa.begin(433E6);

  pinMode(33, INPUT_PULLUP);
  pinMode(15, INPUT_PULLUP);
  pinMode(26, INPUT_PULLUP);
  pinMode(25, INPUT_PULLUP);
}
```

```
void loop()
{

  int value = digitalRead(33);
  int value1 = digitalRead(15);
  int value2 = digitalRead(26);
  int value3 = digitalRead(25);
  Serial.print(value);
  Serial.print(" ");
  Serial.print(value1);
  Serial.print(" ");
  Serial.print(value2);
  Serial.print(" ");
  Serial.println(value3);

  if (value == LOW)
  {
    LoRa.beginPacket();
    LoRa.print("up");
    Serial.println(" up");
    LoRa.endPacket();
    lcd.setCursor(0, 0);
    lcd.print("Camera High");//status
    lcd.setCursor(0, 1);
    lcd.print("Lazer Refrence");//warning
    delay(800);
    lcd.clear();
  }
  else if (value1 == LOW)
  {
    LoRa.beginPacket();
    LoRa.print("down");
    Serial.println("down ");
    LoRa.endPacket();
    lcd.setCursor(0, 0);
    lcd.print("Camera Low");//status
    lcd.setCursor(0, 1);
    lcd.print("Lazer Refrence");//warning
    delay(800);
    lcd.clear();
  }

  lcd.setCursor(0, 1);
  lcd.print("Drone OFF");//warning
  delay(1000);
  lcd.clear();
}
```

Python Coding

```
import cv2
import numpy as np
import time

# Load the neural network model for pose estimation
net = cv2.dnn.readNetFromTensorflow('C:/Users/Akshay ajit/Desktop/openpose/human-pose-estimation-opencv-master/graph_opt.pb')

# Initialize video captur
cap = cv2.VideoCapture(1)

# Specify the input size for the mode
in_width = 368
in_height = 368
threshold = 0.1

# Define the list of keypoints pairs to connect for the body
POSE_PAIRS = [
    (1, 2), (1, 5), (2, 3), (3, 4), (5, 6), (6, 7), (1, 8), (8, 9), (9, 10), (1, 11), (11, 12), (12, 13), (1, 0), (0, 14), (14, 16), (0, 15),
    (15, 17), (2, 17), (5, 16)]

# Minimum number of keypoints for a "complete" posture
MIN_KEYPOINTS = 5
last_detection_time = None
detection_delay = 2
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    # Perform face detection
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.1, 4)

    # Check if any faces are detected
    if len(faces) > 0:
        print("Person found")

    # Prepare the frame for processing
    blob = cv2.dnn.blobFromImage(frame, 1.0, (in_width, in_height), (127.5, 127.5, 127.5),
        swapRB=True, crop=False)
    net.setInput(blob)
    output = net.forward()

    # Print detection message with delay
    current_time = time.time()
    if detected_keypoints >= MIN_KEYPOINTS:
        if last_detection_time is None or current_time - last_detection_time > detection_delay:
            print("Person found")
            last_detection_time = current_time

    # Display the resulting fram
    cv2.imshow('Human Pose Estimation', frame)
    If cv2.waitKey(1) & 0xFF == ord('q'):
        break

    # Release the capture
    cap.release()
cv2.destroyAllWindows()
```

Delay of two seconds so that the pi does not crash if the detection threads/ frequency is high and continues

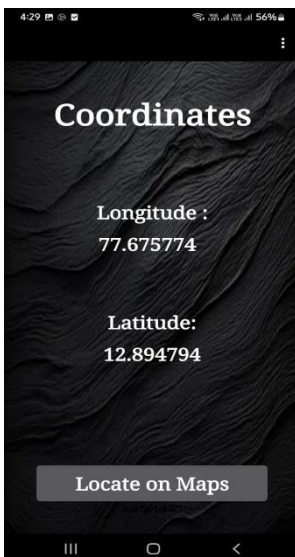
In this code two works are done parrallely detecting face by first converting it in gray scale and also detecting posture by the points and pairs in the body the image is given in computer vision page ahead.

the system will be easy to change or modify by the user and new model is required for different application

User Mobile App

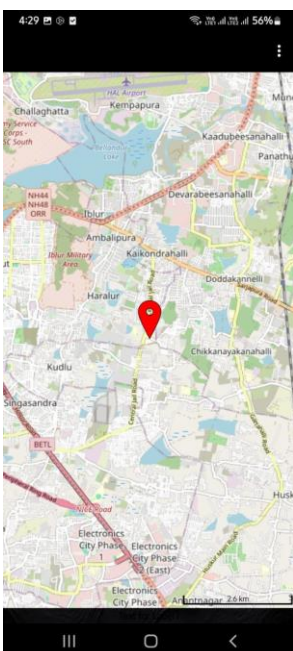


The App starts with a welcome page that depicts the SkyGuard title and the Drone itself. We can add new drone designs depending on the versions very easily. The user will click “Start Detecting” after turning on the Receiver Module. The app then connects itself to receiver to get data and complete task with it.



This App moves to next screen once they click on “start Detecting”. On this page (screen 2) we get the longitude and latitude in digital integer form.

Which can be used by search and recuse or etc personals. The advantage of the app is to reduce the weight of the receiver body on the user but it has to be kept in range

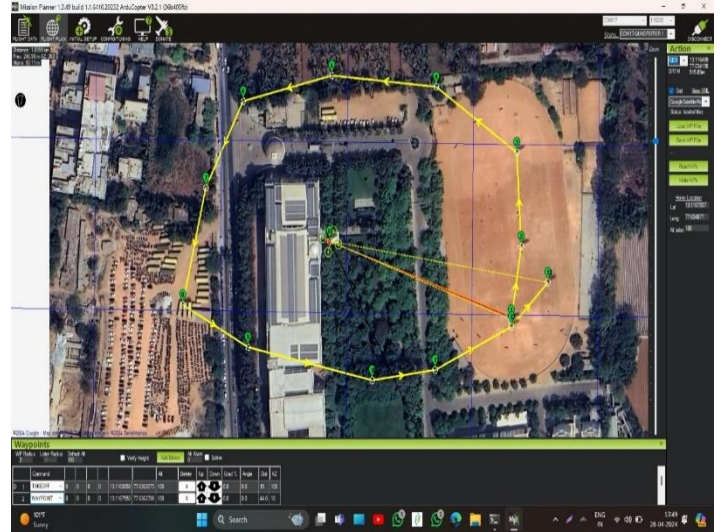


The Maps feature is the best one for the users as they can switch to this page by just clicking on the “Locate on Map” button given on the “Screen 2”.

With this page a user can get a very accurate location relative to space view and also the idea.

In one click it imports the longitude and latitude and directly uses marker at that location.

Mission Planner Software



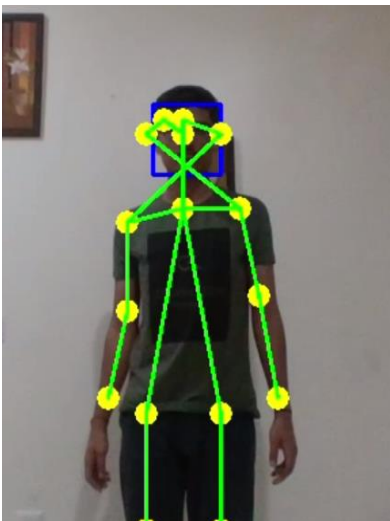
The mission planner waypoint system is a critical component of drone operations, allowing users to plan and execute autonomous flight missions with precision. With this system, users can define specific GPS coordinates, known as waypoints, that the drone will navigate to sequentially.

The mission planner software provides a user-friendly interface for creating and editing waypoints, as well as configuring various flight parameters such as altitude, speed, and loiter time at each waypoint.

Once the mission is planned, the drone can be deployed to execute the mission autonomously, following the predefined flight path. The waypoint system enables drones to perform a wide range of tasks, including aerial surveys, search and rescue missions, and surveillance operations, with minimal human intervention.

Overall, the mission planner waypoint system enhances the efficiency, accuracy, and safety of drone operations by streamlining mission planning and execution.

Computer Vision



Computer vision is a field of artificial intelligence that enables computers to interpret and understand visual information from digital images or videos. By utilizing algorithms and machine learning techniques, computer vision systems can detect objects, recognize patterns, and extract valuable insights from visual data. This technology finds applications in various industries, including healthcare, automotive, surveillance, and robotics, revolutionizing how machines perceive and interact with the world around them.

Over here you can see we are “graph_opt.pb” a pre trained neural model for posture detection in combination with hasscard face detection model both of their accuracy is above 80% as they are trained and improve by a big community which includes our team. We still are hoping to add hand detection separately to increase accuracy even further.

Computer Vision

What is the graph_opt.pb file?

- The graph_opt.pb file contains the weights of a pre-trained neural network model.
- Specifically, it is associated with human pose estimation tasks.
- These weights represent the learned parameters of the neural network, which have been fine-tuned during training.

How Does It Work?

- When performing human pose estimation, the neural network (often based on architectures like OpenPose or MobileNet) processes input images or video frames.
- The network analyzes the visual features and identifies key points corresponding to human body joints (e.g., shoulders, elbows, hips).
- The graph_opt.pb file provides the necessary information for the neural network to determine the human's position and orientation (pose).
- By leveraging the learned weights, the network predicts the joint locations, which can then be used to reconstruct the complete human pose.

Integration with OpenCV:

- In OpenCV-based applications, the graph_opt.pb file is loaded alongside the neural network architecture.
- The neural network processes input data (such as frames from a camera feed) and produces pose estimations.
- These estimations can be used for applications, including gesture recognition.

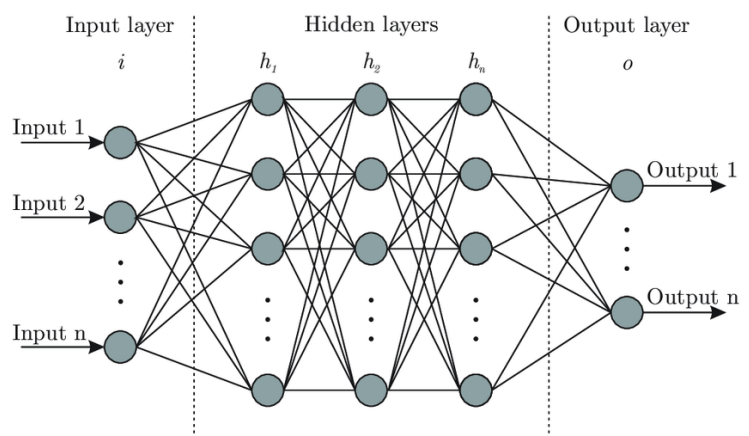
The future models to explore:

- Currently we are working on developing models for pothole detection that is a big issue in Indian roads having a list of potholes by municipal corporations will help in solving the issues .
- We are trying to improve our pair detection in posture software it sometimes matches the wrong pair but with proper training and refined code it will be more accurate
- The app that users have should allow the user to switch between detection from posture to pothole which can be very useful in the terms of modularity.

Neural networks:

A neural network is a computational model inspired by the structure and functioning of the human brain's interconnected network of neurons. It's a key component of deep learning, a subset of machine learning. Neural networks consist of interconnected nodes, called neurons, organized in layers

there are nodes and so called neurons in the network it is important for neural network its base of ai and ml based learning



A Neural Network

Calculations

Thrust Calculation

Motor power : 80% Throttle

Battery : 12volts / 3300mah

ESC temperature : optimum 45C

Thrust : Calculated using a weight machine.

Thrust : 1.4 kg/ each

Thrust : 13.72931 Newton/ each

Total 4 Motor Thrust : 5.6 kg

Total 4 Motor Thrust : 54.91724N

Power of each motor: 150watts(Max)

Thrust to weight ratio : 2.8(approx)

Fly Time

Motor power : 80% Throttle

Battery : 12volts / 3300mah

ESC temperature : optimum 45C

Total 4 Motor Thrust : 5.6 kg

Total current consumption : 15Amp(peak)

Total payload weight : 1kg (AAD)

Total Amps : 20

Battery amps : 3500

Fly time unit :

Time = capacity × discharge / AAD

700 = 3500 × 20 / 100

So 700 sec is 12 mins(approx.)

Payload carrying capacity

Motor power : 80% Throttle

Battery : 12volts / 3300mah

ESC temperature : optimum 45C

Total 4 Motor Thrust : 5.6 kg

Total drone weight with attachments : 1.803kg (2kg approx.)

Extra weight carrying capacity :

Total thrust – Total drone weight :

5.6 kg – 2.0kg = 3.6 kg

Total carrying capacity Capacity < 3.6kg

Applications

Border Security: SkyGuard enhances border security by autonomously patrolling and monitoring border regions. Its advanced sensors and computer vision technology detect and track unauthorized crossings people who might be smuggling drugs or cows inside the border, providing real-time alerts to border patrol agents.

Search and Rescue: In search and rescue missions, SkyGuard scans vast areas to locate missing individuals. Its rapid deployment, autonomous flight capabilities, and precise person detection algorithms expedite rescue operations, saving valuable time and resources.

Wildfire Detection: SkyGuard detects wildfires early by continuously monitoring forested areas. Its thermal imaging and advanced sensors identify heat signatures, enabling timely intervention to mitigate fire spread and prevent disasters.

Pothole Geotagging: Utilizing its onboard sensors and GPS technology, SkyGuard conducts aerial surveys to geotag potholes on roads. This data facilitates efficient road maintenance and repair efforts, improving transportation infrastructure and ensuring road safety.

Building Inspection: SkyGuard conducts aerial inspections of buildings and structures, capturing high-resolution images and identifying potential structural defects or safety hazards. Its versatility and precision make it an invaluable tool for building maintenance and safety assessments.

Wild Life Protection: SkyGuard systems can be trained to detect a human who can be a poacher roaming in forests to hunt animals like lion rhino to hurt them for their valuable. The system can sweep the areas continuously. if a poacher is found or detected it will send a geo location that can be used by authority to track the poacher down . it can also be used to understand and research about animals migration pattern using multiple models which are custom built.

Market Analysis

Madan sir : “a good project should also have a market demand or it does not have role in changing the world”

Spectrum For Karnataka

Karnataka :

Number of Fire Stations 152

Interested Market 50 (least assumption)

Number of Industrial Parks 174

Interested industrial parks 60 (least assumption)

Number of High Security Facilities 4023

Interested to Acquire 200 (least assumption)

Build cost 22,000/- can be sold for 30,000/-

Yearly Maintenance and Software fess 5000/-

Turn Over : 93,00,000/-

Yearly Fees Charges : 15,50,000/-

Total Turn Over : Rupees 10,850,000/- (Approx)

Spectrum For India

India:

Data Yet to crunched as the numbers are big and state wise demand differs we have a rough number but it is yet to be vetted and analysed

The opportunity will obviously be 3 times higher than Single state

Interaction with Experts/Future Customers



Mr. Lakshmi Prasad

Deputy Commissioner of
Police (DCP),
North East Section,
Bangalore, Karnataka, India

“The Project SkyGuard Made By the Students. will help us in patrolling activities. If reinforces our capabilities by adding strength in Rescuing people”



Mr. Anil Digambar Ursal

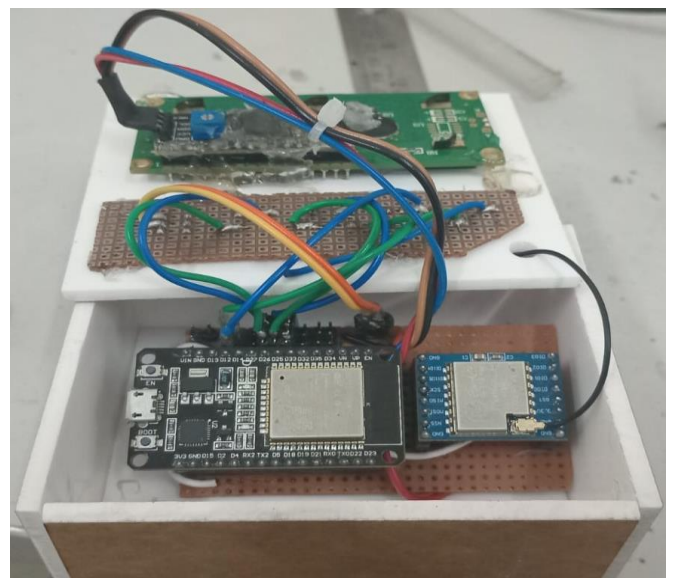
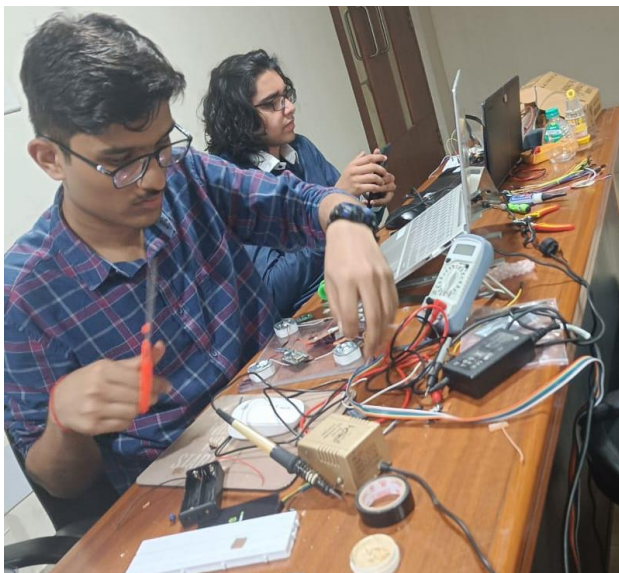
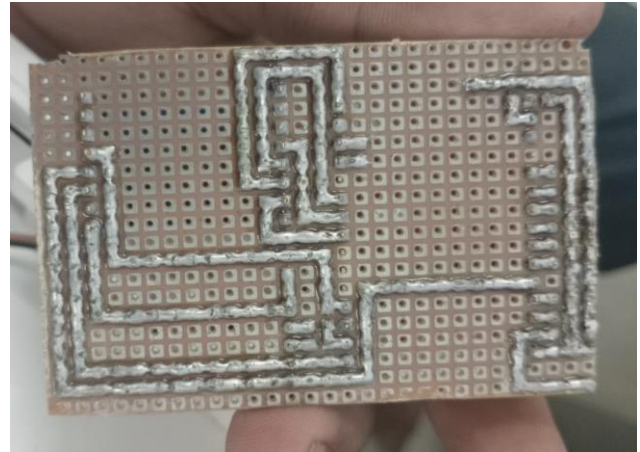
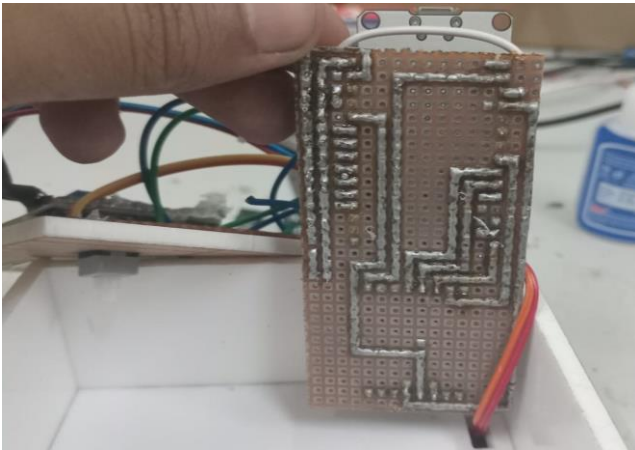
Sub – Inspector (SI),
South Section, Mumbai,
Maharashtra, India

“SkyGuard is a great innovation made by the students and will surely help us in securing restricted area and government lands and properties. I can see its potential in border patrol near India-Pakistan from where narcotics has been smuggled by human carriers”

Safety Features

- **Motor and Battery Monitoring:** Continuous monitoring of motor and battery health to prevent overheating or malfunction.
- **Auto-Landing:** Automatic landing protocol activated in case of drone disconnection or critical system failure.
- **Altitude Hold:** Flight controller equipped with altitude hold functionality for stable flight even in adverse conditions.
- **Speed, Altitude, and Acceleration Limits:** Built-in limits to prevent the drone from exceeding safe speed, altitude, or acceleration thresholds.
- **Range Limits:** Predefined boundaries to restrict the drone's flight range, ensuring it stays within a safe operating area.
- **Secondary GPS:** Redundant GPS system for backup navigation in case of primary GPS failure or signal loss.
- **Fail-Safe Protocols:** Fail-safe mechanisms programmed to activate in case of system errors or communication disruptions.
- **Real-Time Monitoring:** Continuous real-time monitoring of vital system parameters and feedback for early detection of anomalies.
- **Manual Enable/Disable Functionality:** The pilot can manually enable or disable safety features as needed, providing flexibility in operation while ensuring adherence to safety protocols.

Fabrication



The Dedicated & Disciplined team has a very diverse talents.2 are great in hardware & 2 are great in software. We together are able to achieve amazing Goals. And work more as a Family then a Team. With a strong bond from last 3 years and counting. Together we solve problems and contribute to the community as much as possible in our powers. We share our knowledge to community via free Workshops and Competitions in our University

Challenges Faced & Solved

1. Long range data transition?

Solved : we were having issues with deciding how to send the coordinates we had an easy idea of Bluetooth but it will obviously be very short range, then we thought WiFi but it won't be available in remote area where the drone is supposed to be. This problem helped us to get to LoRa module this board is able to use radio frequencies to communicate in less time or latency while keeping the overall device light weight and simple

2. Construction materials?

Solved : we faced issues of selecting a proper material to make this design. first we thought that we can do aluminium custom parts. But after brainstorming and thinking about it we got Acrylic as a best candidate it might not be as strong as aluminium but the load of the drone was less anyway after stress test we settled to Acrylic sheet 5mm and 3mm. white and black which is contrast to the background sky.

3. Landing procedure for beginner?

Solved : the landing and throttle control for a beginner in stable mode is hard and very tricky to solve this issue we implemented the land mode which automatically lands the drone on the ground without user interaction with throttle the user anyway has autonomous mode to do all the process without interaction from takeoff to landing all can be planned and programmed in Auto mode

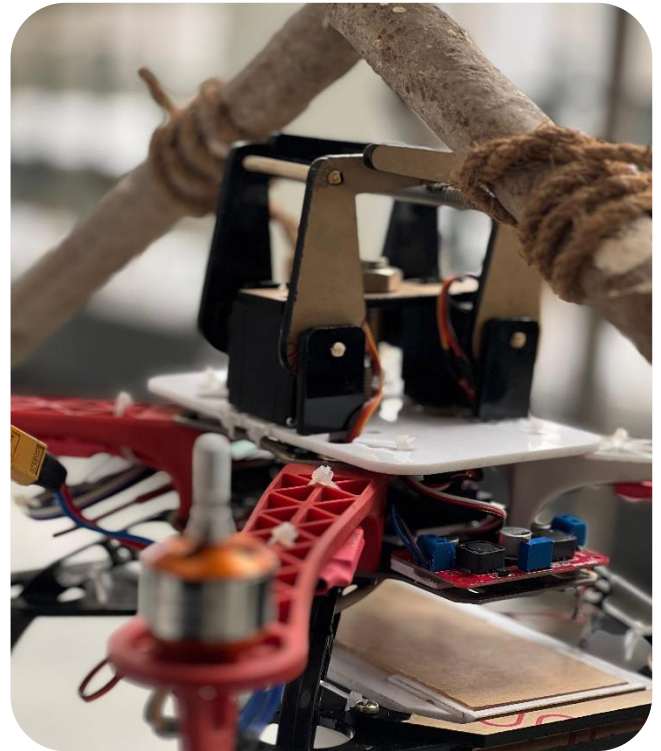
4. Issues with False Detection

Solved : the system was very much inaccurate and was detecting random objects we thought about this and implemented 3 changes to remove it totally. First we implemented using models that are not only developed by us but is also improved by the community around us. Then we also implemented double feature check in which it detects posture but also tries to find and detect face for a secondary conformation. Then we switched from the bad quality of Webcam to Pi camera with better Resolution and clarity. These steps removed the issue of inaccuracy and false alarm

Results



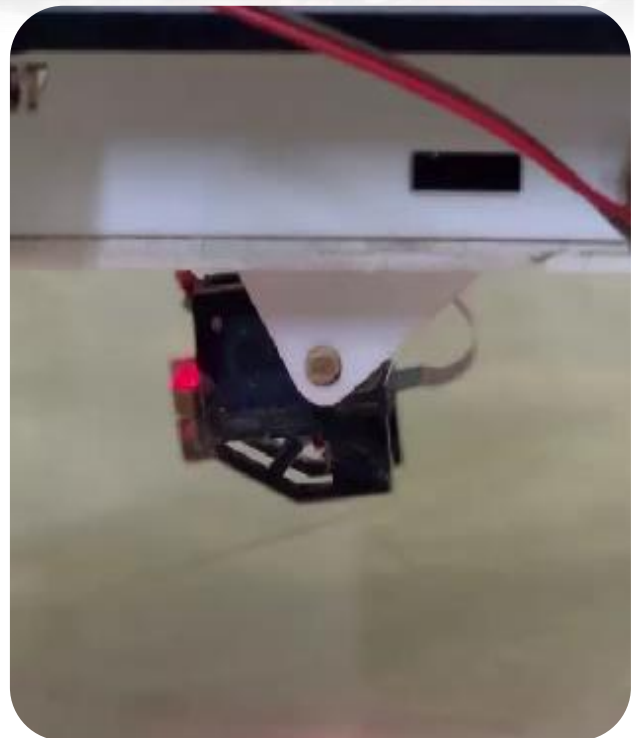
The Flight



The Gripper



Location Sending



Camera Tilt

Conclusion

In conclusion, the integration of Raspberry Pi 4 with Camera Module, ESP32 Microcontroller with GPS Sensor, and LoRa Transceiver Modules in conjunction with Ardupilot Flight Controller and RC Controller represents a holistic approach to drone technology. This synergy of hardware components and software systems empowers the SkyGuard project with unparalleled capabilities in surveillance, navigation, and communication.

The Raspberry Pi 4 serves as the computational backbone, leveraging its robust computing capabilities and compatibility with peripherals to capture and process real-time imagery crucial for person detection. Coupled with TensorFlow for advanced machine learning algorithms, it enables accurate identification and tracking of individuals within the monitored area.

The ESP32 Microcontroller, equipped with a GPS Module, adds another layer of functionality by providing precise location data essential for transmitting the user's coordinates to remote monitoring stations. Its low power consumption and wireless connectivity ensure efficient communication with the Raspberry Pi 4 and other devices, facilitating seamless integration into the SkyGuard system.

Together, these components form a robust and versatile platform that can be tailored to meet the unique requirements of various applications. Whether deployed in military border patrols, disaster management, or industrial inspections, SkyGuard embodies innovation and adaptability, empowering users to enhance safety, security, and efficiency in their operations.

In summary, the SkyGuard project represents a convergence of cutting-edge technologies aimed at addressing real-world challenges and advancing the capabilities of drone systems. With its comprehensive feature set and customizable design, SkyGuard stands poised to revolutionize aerial surveillance and exploration across diverse sectors and applications

-Thank you